

SHANMUGARAJA KAVEEN

Software Engineering Undergraduate

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Professional Summary

Motivated Software Engineering undergraduate at Sabaragamuwa University of Sri Lanka (GPA: 3.58/4.00), seeking a Software Engineering internship. Experienced in full-stack and AI-powered application development using React, Next.js, Node.js, FastAPI, and Python. Skilled in building RAG-based systems, RESTful APIs, and scalable database-driven applications. Developed cross-platform mobile apps with Firebase and led a university-level AI research platform project. Passionate about software architecture, clean code, and building production-ready systems.

Education

Bachelor of Science (Hons) in Software Engineering <i>Sabaragamuwa University of Sri Lanka</i>	2023 – 2027 (Expected) <i>GPA: 3.58 / 4.00</i>
G.C.E Advanced Level <i>Badulla Central College, Sri Lanka</i>	2021 / 2022

Relevant Coursework

Data Structures & Algorithms | Object-Oriented Programming | Database Management Systems | Software Engineering | Web Application Development | Software Verification & Validation

Work Experience

Freelance Software Developer <i>Self-Employed</i>	2024 – Present
- Developed and maintained responsive full-stack web applications using React, Node.js, and MongoDB. - Integrated RESTful APIs and implemented scalable frontend-backend communication workflows. - Improved application performance, responsiveness, and cross-browser compatibility for production-ready deployments.	
Team Lead — FutureHive Capstone Project <i>Sabaragamuwa University of Sri Lanka</i>	2025
- Led development of a full-stack AI-powered research assistant platform. - Managed Agile workflows, Git collaboration, feature planning, and system architecture. - Coordinated frontend and backend integration while ensuring scalable application structure.	

Technical Skills

Programming: Python, JavaScript, Java
Frontend: React, Next.js, React Native, Flutter, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, Django, Flask
Databases: MongoDB, Firebase, ChromaDB
Software Engineering: RESTful APIs, Object-Oriented Programming (OOP), MVC Architecture, Agile/Scrum, Database Design
AI/ML: RAG Pipelines, LangChain, Ollama, OpenAI API
Tools: Git, GitHub, Postman, VS Code

Projects

FutureHive — AI-Powered Research Capstone Project Assistant  <i>React, FastAPI, MongoDB, ChromaDB, Ollama (LLaMA 3)</i>	2025
- Developed an AI-powered platform that helps students explore past research papers, discover trending research topics, and receive guidance for research and capstone projects. - Built an intelligent chatbot that answers user questions based on selected research papers using Retrieval-Augmented Generation (RAG). - Implemented semantic search and document retrieval pipelines using ChromaDB for efficient access to research paper	

- content.
- Developed the full-stack application using React, FastAPI, and MongoDB with scalable RESTful APIs.
 - Integrated Ollama (LLaMA 3) to provide contextual question answering and intelligent research recommendations.
 - Applied modular system architecture and Agile development practices to support future scalability and feature expansion.

Saree Bazar — E-commerce Platform with Virtual Try-On

2026

Next.js, Node.js, MongoDB, IDM-VTON

- Developed a full-stack e-commerce platform that enables customers to browse, purchase, and virtually try on sarees before placing orders.
- Built the application using Next.js, Node.js, and MongoDB with a scalable and maintainable architecture.
- Implemented responsive UI components and optimized product filtering and checkout workflows.
- Integrated AI-powered virtual try-on functionality using IDM-VTON to enhance customer engagement and shopping confidence.
- Applied Framer Motion animations and responsive design principles to improve overall user experience.
- Optimized SEO, application performance, and cross-browser compatibility for production deployment.

Memora — Memory Mapping and Social Gallery Mobile Application

2026

React Native, Firebase, Firestore, Cloud Storage

- Developed a cross-platform mobile application that allows users to store, organize, and share memories through photos, location-based maps, and social interactions.
- Built the application using React Native with Firebase Authentication, Firestore, and Cloud Storage for secure data management.
- Implemented interactive map features that connect uploaded memories to their geographical locations.
- Developed social networking features including real-time chat, follow, like, comment, and share functionality similar to social media platforms.
- Designed responsive mobile interfaces and cloud-based image storage for a seamless user experience.
- Applied privacy controls and access management mechanisms to ensure secure user interactions and content sharing.

Leadership & Activities

Team Lead — University Community Project

2026

Sabaragamuwa University of Sri Lanka

- Led a student team to successfully plan and deliver a university community project.
- Managed task allocation, deadlines, and team coordination to ensure successful completion.

Hackathon Participant — PearlHack 2025

2025

IEEE WIE Affinity Group, Sabaragamuwa University of Sri Lanka

- Collaborated with a team to design and develop a software solution within a limited timeframe.
- Gained hands-on experience in rapid prototyping and Agile-based software development practices.

Languages

English | Tamil | Sinhala

References

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